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Tompane

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(54) **FRAME ARRANGEMENT FOR WOOD FRAMED BUILDINGS**

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(51) **Int. Cl.**

E04B 1/26	(2006.01)
E04B 2/70	(2006.01)
E04B 5/12	(2006.01)
E04B 7/02	(2006.01)

(52) **U.S. Cl.**

CPC ***E04B 5/12*** (2013.01); ***E04B 1/2612*** (2013.01); ***E04B 2/70*** (2013.01); ***E04B 7/022*** (2013.01)

(58) **Field of Classification Search**

CPC . E04B 1/2612; E04B 2/70; E04B 5/12; E04B 7/022

USPC 52/633

See application file for complete search history.

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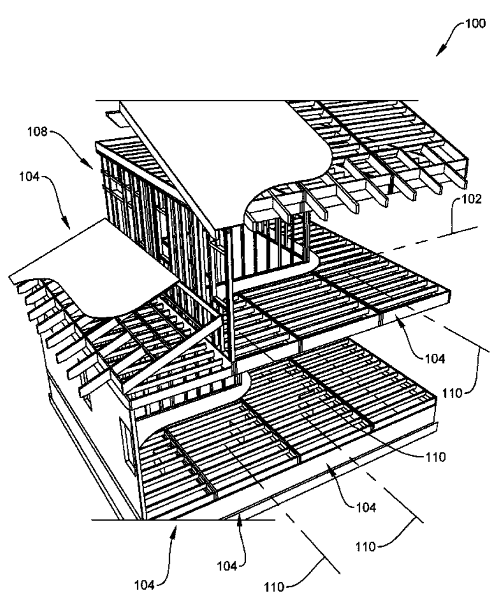
(57) **ABSTRACT**

A method of constructing a frame of a wood frame building structure is shown and described. The method entails fabricating modules establishing nominally flat structural platforms for expediting assembly of floors, walls, ceilings, and a roof. These modules provide structural frames enabling appropriate finishing materials to be affixed thereto. The finishing materials provide continuous, flat surfaces, with exceptions for penetrations. The method includes assembling the modules such that a longitudinal axis of each model is parallel to those of other modules throughout the frame and perpendicular to a building length axis. Also, each module includes ceiling joists, floor joists, and roof rafters, as appropriate, such joists and roof rafters installed parallel to the building length axis and therefore, perpendicular to the joists and roof rafters. This method minimizes assembly costs of modules and of the framing, while retaining conventional framing performance.

4 Claims, 5 Drawing Sheets



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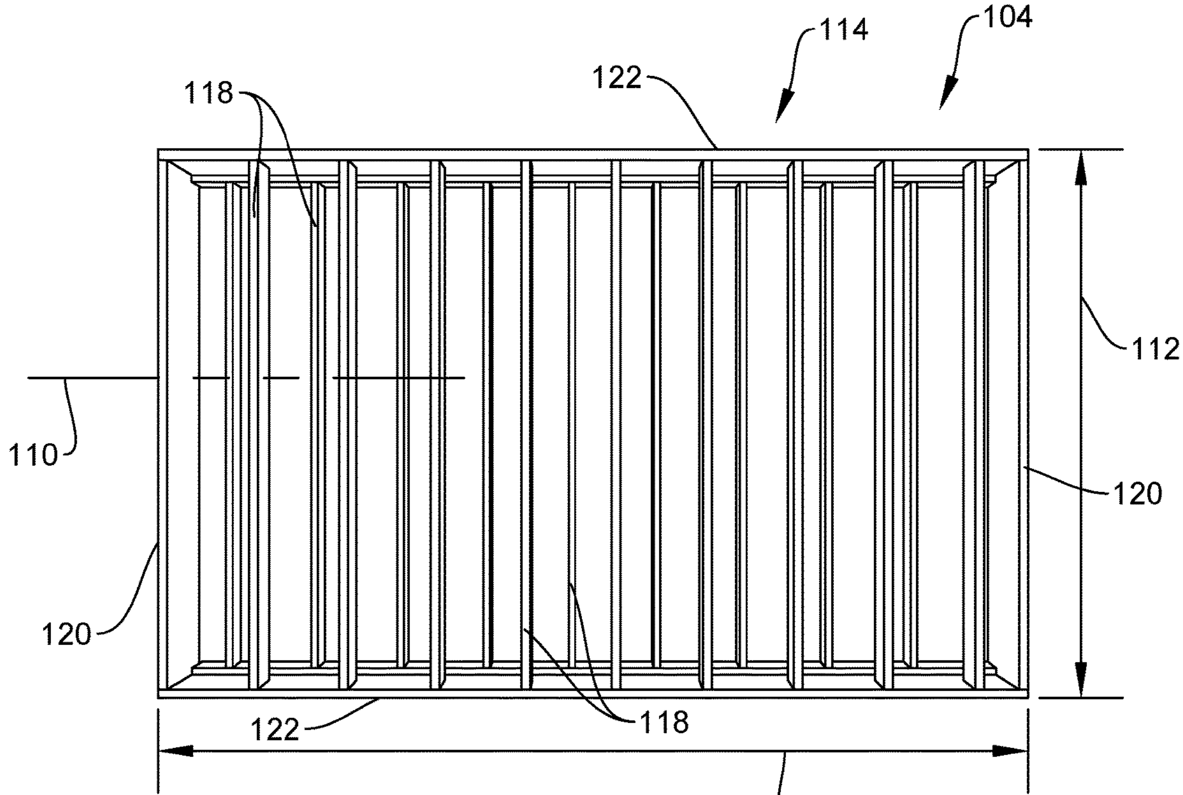
114
104
122
118
112
110
120
120
122
118
130

FIG. 1

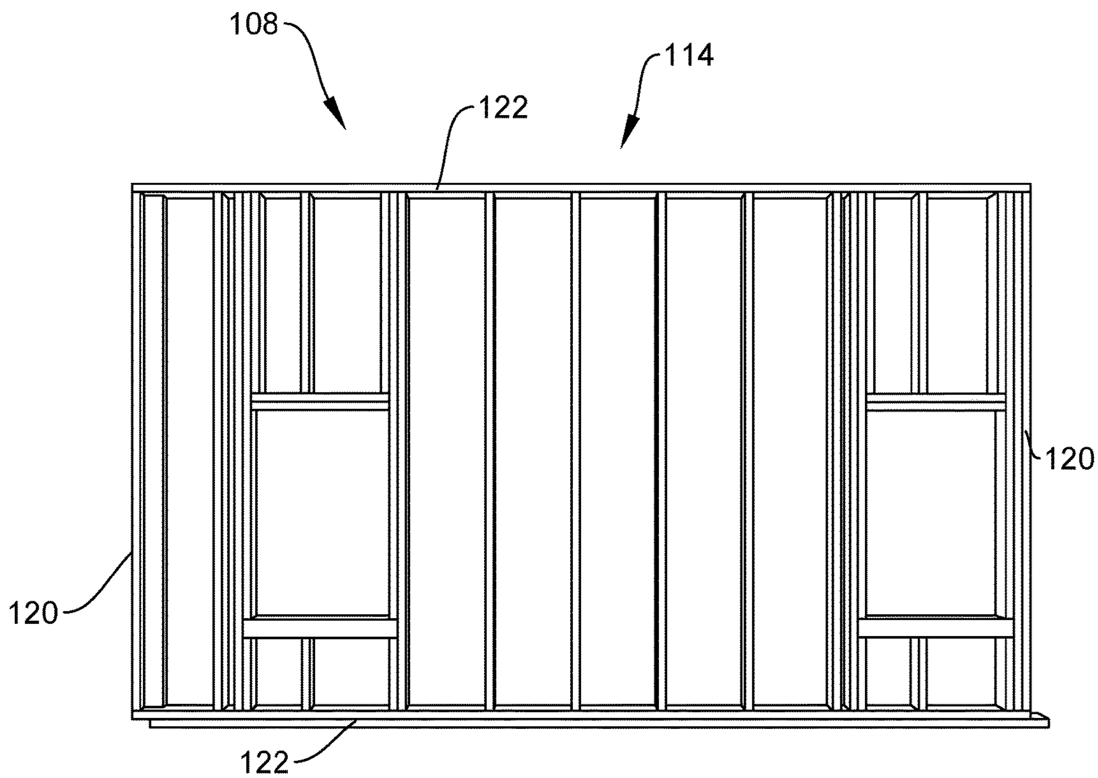
108
114
122
120
120
122

FIG. 2

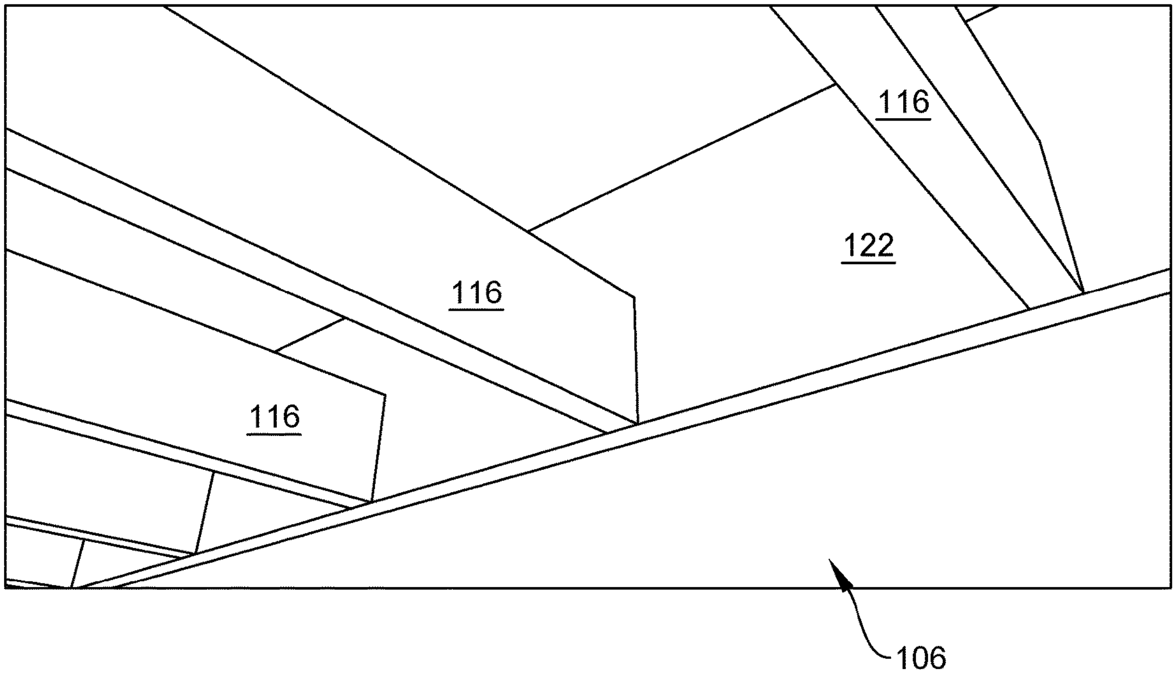

FIG. 3

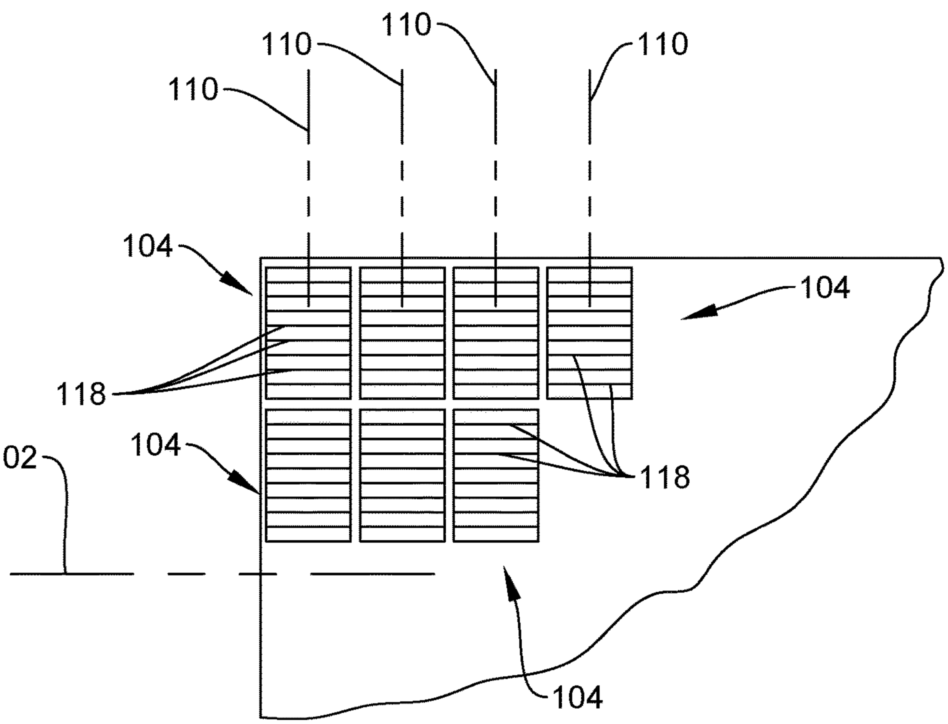

FIG. 6

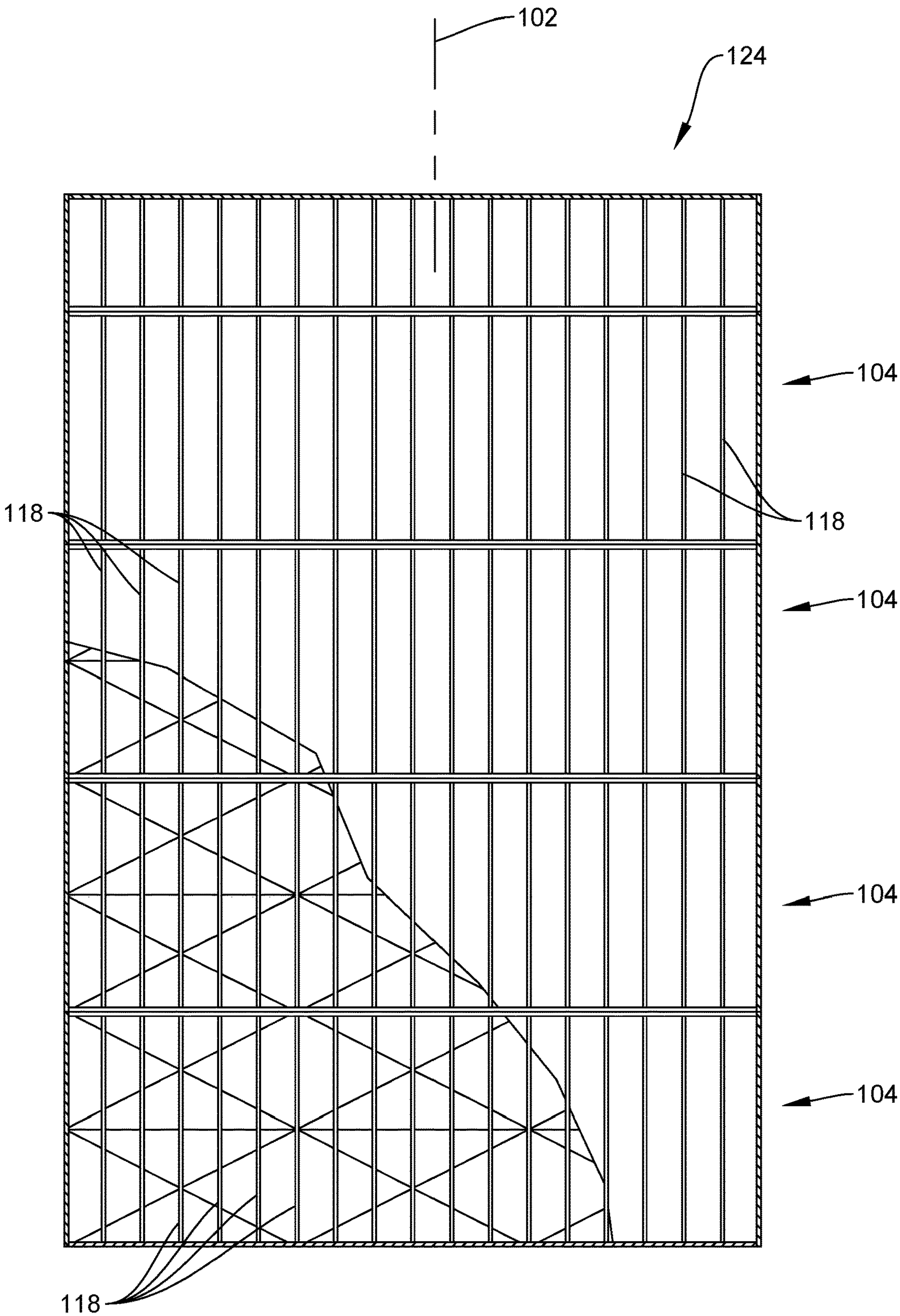

FIG. 4

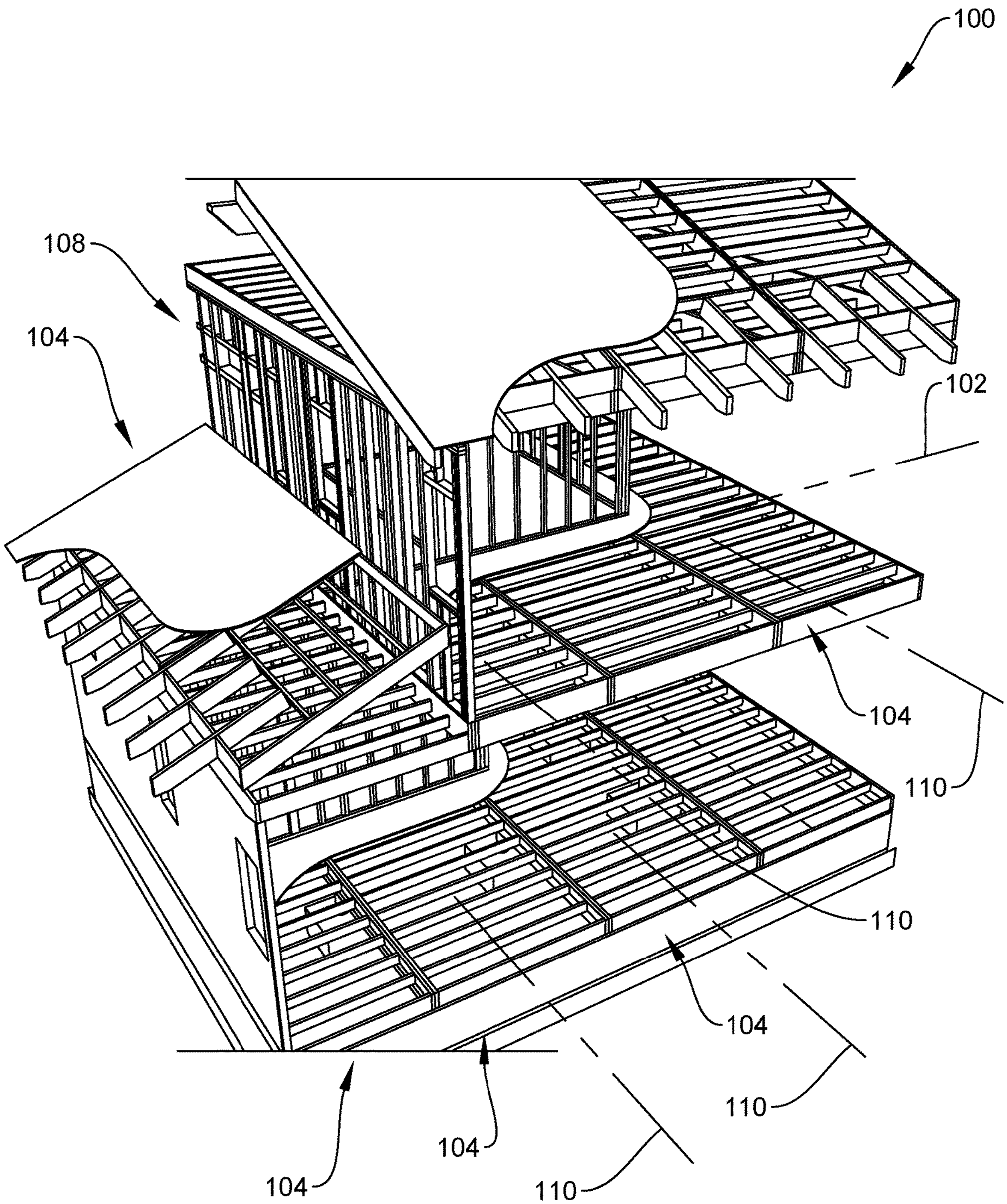

FIG. 5

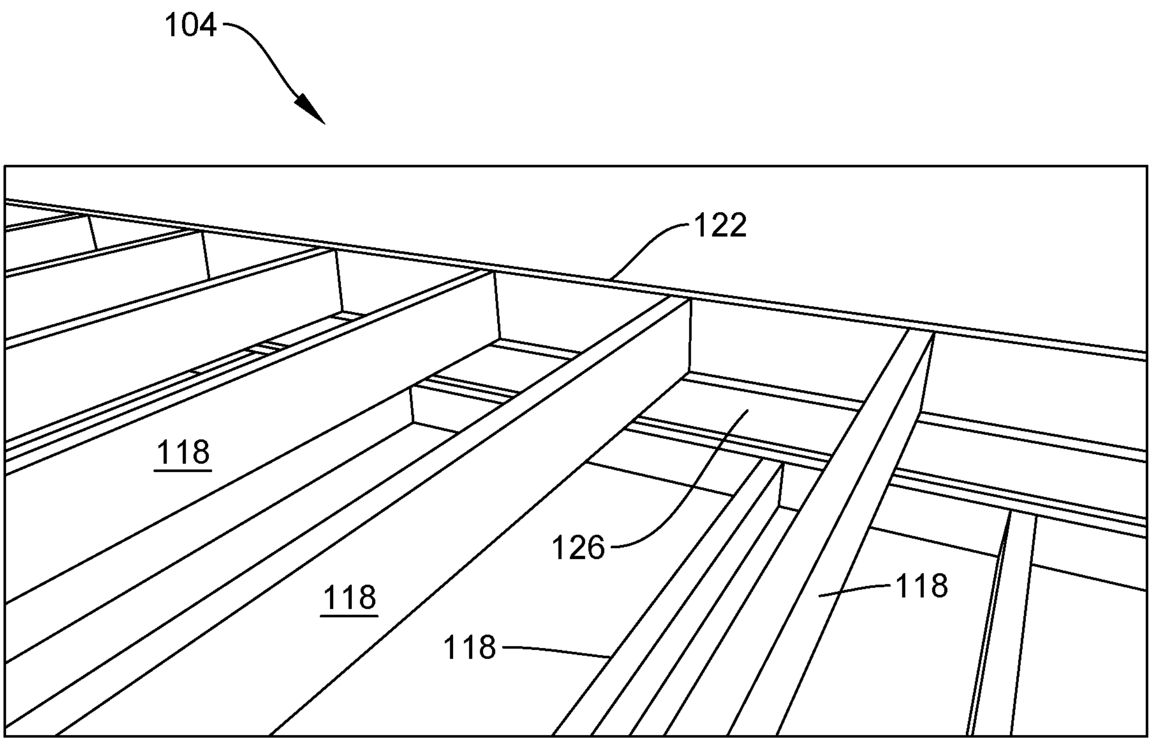

FIG. 7

FRAME ARRANGEMENT FOR WOOD FRAMED BUILDINGS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application relies on the disclosure of and claims priority to and the benefit of the filing date of U.S. Provisional Application Nos. 63/109,525, and 63/109,528 both of which were filed on Nov. 4, 2020, the disclosure of both is hereby incorporated by reference herein in their entirety. This application is also a Continuation in Part of U.S. Non-Provisional patent application Ser. No. 17/394,737, filed Aug. 5, 2021, and is also hereby incorporated by reference.

FIELD OF THE INVENTION

The present invention relates to construction of wood frame buildings, and more particularly to modules finding utility in building wood frames for buildings.

BACKGROUND OF THE INVENTION

Construction of wood framed buildings is most commonly performed by cutting and coupling individual boards at a construction site. It is desirable to minimize labor in building construction because labor is one of the most expensive inputs in the construction process. Additionally, using human labor generates a certain amount of waste of material.

One answer to the issue of economy is the use of prefabricated modular panels, or modules. This practice is frequently used for certain generally irregular shaped components, such as triangular trusses for supporting a roof. Such components are generally large and labor intensive to build by hand Reliance on modules for building extensive platforms such as floors, ceilings, and roofs has not met with widespread commercial success.

Greater economies than currently practiced could be realized if there were practical ways to utilize modules for extensive platforms.

SUMMARY OF THE INVENTION

The present invention addresses the above stated need by proposing advantageous schemes to incorporate modules into extensive platforms, and by proposing compliant modules arranged advantageously for automated assembly procedures.

In one aspect, the present invention sets forth a method of constructing a frame of a wood frame building structure, the method enabling use of expeditiously fabricated modules. The method comprises fabricating modules establishing nominally flat structural platforms for expediting assembly of floors, walls, ceilings, and a roof. These modules provide structural frames enabling appropriate finishing materials to be affixed thereto. The finishing materials provide continuous, flat surfaces, with exceptions for penetrations such as doors, windows, access passages for heating, cooling, and ventilating ducts, plumbing, wiring, and the like.

Modules enable expedited assembly of the building frame, while retaining conventional advantages of otherwise conventional wood framing.

Beyond merely using modules, the present invention contemplates assembling the modules such that a longitudinal axis of each model is parallel to those of other modules throughout the frame and perpendicular to a building length axis. Also, each module includes ceiling joists, floor joists, and roof rafters, as may be appropriate, such joists and roof rafters installed parallel to the building length axis and therefore, perpendicular to the joists and roof rafters.

This arrangement contributes to structural integrity of the building frame, while accommodating expeditious and advantageous assembly of modules into the building frame, and enables economies to be realized when building the modules in a facility remote from the building being constructed.

It is anticipated that standardizing layout of wall, floor, and roof modules as to top and bottom plates, and joists or rafters, costs of machinery, manpower, and factory spaced may be reduced by a factor of three, allowing additional savings over module costs set forth above. It should further be observed that the novel assembly concept enables modules to be built using a minimum of different lumber sizes. This reduces inventory and storage costs to the module manufacturer.

Another benefit is that of improved sound proofing. It is anticipated that whereas a conventionally framed and insulated wall or floor system receives an expected Sound Transmission Class (STC) rating of 38, using novel modules where conventional twelve inch wide joists or roof rafters would be used yields an STC rating of 58.

Still additional benefits include improved thermal insulation performance. Because joists in the novel modules are provided in two vertically separated sections rather than one continuous joist (e.g., two two inch by six inch joists rather than one two inch by twelve inch joist), and with horizontal spacing as well, thermal and acoustic performance both increase.

The present invention provides improved elements and arrangements thereof by apparatus for the purposes described which is inexpensive, dependable, and fully effective in accomplishing its intended purposes.

These and other objects of the present invention will become readily apparent upon further review of the following specification and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

Various objects, features, and attendant advantages of the present invention will become more fully appreciated as the same becomes better understood when considered in conjunction with the accompanying drawings, in which like reference characters designate the same or similar parts throughout the several views, and wherein:

FIG. **1** is a plan view of one type of module used in the present method and building;

FIG. **2** is a plan view of a second type of module used in the present method and building;

FIG. **3** is a plan view of a third type of module used in the present method and building;

FIG. **4** is a plan view of a complete deck of a building built by the present method;

FIG. **5** is a perspective view of a building frame supported on a foundation, partially broken away to reveal internal detail;

FIG. **6** is a schematic partial plan view of a deck of a building; and

FIG. **7** is a top perspective detail view of a variant of the module of FIG. **1**.

Drawings **1**-**7** are drawn to internal scale, but not necessarily to external scale. By internal scale it is meant that the parts, components, and proportions thereof in the illustrated inventive example are drawn to scale relative to one another. As employed herein, external scale refers to scale of the illustrated example relative to scale of environmental elements or objects, regardless of whether the latter are included in the drawings. Where the inventive example claims external scale, the inventive and environmental elements may of course not be drawn to real or true life scale; rather, external scale signifies only that both the invention and environmental elements are drawn in scale to each other.

DETAILED DESCRIPTION

Referring first to FIGS. **1**-**3**, according to at least one aspect of the invention, there is shown a method of constructing a frame **100** (FIG. **5**) of a wood frame building structure (not shown in its entirety) having a building length axis **102**. The method enables use of expeditiously fabricated modules **104**, **106**, **108**. The method comprises fabricating a plurality of modules **104**, **106**, **108** including floor or roof modules **104**, ceiling modules **106**, and wall modules **108**. Each module **104**, **106**, or **108** comprises a module length axis **110**, a module width **112**, a bounded wood perimeter **114**, and ceiling joists **116**, floor joists **118** or roof rafters **118** oriented perpendicularly to length axis **110**. The method further comprises building framing **100**, and assembling frame **100** of the wood frame building structure using modules **104**, **106**, and **108**, wherein floor modules **104**, ceiling modules **106**, and wall modules **108** are coupled to the building wall framing **100** with respective module length axes **110** perpendicular to building length axis **102**. Ceiling joists **116**, floor joists or roof rafters **118** are parallel to building length axis **102**.

The relationship of modules **104**, **106**, **108** to frame **100** of the building structure is shown for emphasis in FIG. **6**. All modules **104**, **106**, **108** (represented by module **104** in the example of FIG. **6**) are oriented with module length axes **110** perpendicular to building length axis **102**. Consequently, joists and rafters (represented by joists **118**) are parallel to building length axis **102**. A more literal rendering is seen in FIG. **4**, the latter showing a completed deck **124**. Deck **124** is a flat surface having a depth extending vertically, and accounts for a footprint area of the building under construction, e.g., a first story floor, a second story floor, etc.

Bounded wood perimeter **114** comprises end boards **120** and top and bottom boards **122**. End boards **120** and top and bottom boards **122** collectively establish perimetric bounds of all modules **104**, **106**, **108**. An exception is modification to prefabricated modules **104**, **106**, **108** to accommodate a misfit between module size and remaining area of a frame **100** remaining to be built. This situation occurs when a floor plan has a footprint area that does not jibe with area coverage of a whole number plurality of modules **104**, **106**, **108**, and a portion of frame **100** must be custom built.

Floor and roof modules **104** are structurally and functionally similar, and therefore share a common reference numeral. Because of this, both joist boards and roof rafters are designated with the singular reference numeral **118**.

In the method, the step of building wall framing **100** comprises building at least a portion of building wall framing **100** from rectangular wall modules **108** each including a bounded wood perimeter **114** and having a longitudinal wall module axis **110** and structural timbers arranged perpendicularly to longitudinal wall module axis **110**, and coupled to bounded wood perimeter **114**. Alternatively stated, constructing only part of a building using the novel method is regarded as within the scope of the inventive method.

In the method, each module **104**, **106**, or **108** abuts another module **104**, **106**, **108** when installed to frame **100**. This is shown in FIG. **5**. FIG. **6** is an exaggerated view showing modules **104** slightly separated only for clarity of understanding.

In the method, floor modules **104** and roof modules **104** comprise upper joist boards **118** and lower joist boards **118** parallel to upper joist boards **118** and lower joist boards **118** of every other respective said floor module **104** or roof module **104**. This is illustrated in the perspective view of FIG. **1**. Further description of upper and lower joist boards **118** may be obtained from the copending application referenced at the beginning of this application.

In the method, at least one module **104**, **106**, or **108** may optionally include at least one ledger board **126**. This is shown in FIG. **7**, where ledger board **126** supports upper joists **118**.

In the method, the step of fabricating a plurality of modules **104**, **106**, **108** may further comprise fabricating floor modules **104** and roof modules **104**, the ceiling modules **106**, and the wall modules **108** to be identical in length **130** (shown for floor and roof module **104** in FIG. **1**) and width **112** (FIG. **1**).

In the method, the step of fabricating a plurality of modules **100**, **106** further comprises fabricating floor modules **100**, roof modules **100**, and ceiling modules **106** to be identical in size and in spacing of floor joists **118**, roof rafters **118**, and ceiling joists **116**.

Referring to all Drawing FIGS., the invention may be thought of as a wood frame building structure having a wood frame **100** and a building length axis **102**. The wood frame building structure may comprise a plurality of floor modules **104** or roof modules **104** coupled to wood frame **100**, and a plurality of ceiling modules **106** coupled to wood frame **100**. Each floor module **104** has module length axis **110** oriented perpendicularly to building length axis **102**, bounded wood perimeter **114**, and floor joists **118** oriented perpendicularly to module length axis **110**. Each roof module **104** has module length axis **110** oriented perpendicularly to building length axis **102**, bounded wood perimeter **114** and roof rafters **118** oriented perpendicularly to module length axis **110**. Ceiling module **106** has module length axis **110** oriented perpendicularly to building length axis **102**, bounded wood perimeter **114** and ceiling joists **116** oriented perpendicularly to module length axis **110**. Each floor module **104** abuts another floor module **104** when installed to frame **100**. Each ceiling module **106** abuts a floor module **104** when installed to frame **100**. Each roof module **104** abuts another roof module when installed to frame **100**. Floor modules **104**, roof modules **104**, and ceiling modules **106** are identical in length **130** and width **112**, and are identical in spacing apart of floor joists **118**, roof rafters **118**, and ceiling joists **116**.

In the wood frame building structure, floor modules **104** and ceiling modules **106** comprise upper joist boards **118** or **116** and lower joist boards **118** or **116** parallel to upper joist boards **118** or **116** and lower joist boards **118** or **116** of every other respective floor module **104** or ceiling module **106**. Upper and lower joist boards **118** or **116** are spaced apart vertically, to generate an open space therebetween. This open space enables ready installation of heating, air conditioning, and ventilating ducts, plumbing, and wiring in floors and ceilings.

Optionally, in the wood frame building structure, at least one module **104**, **106**, or **108** includes at least one ledger board **126** (FIG. **7**) supporting at least one other structural component of the wood frame building structure.

It should be noted at this point that orientational terms such as vertically, top, bottom, upper, and lower refer to the subject drawing as viewed by an observer. The drawing figures depict their subject matter in orientations of normal use, which could obviously change with changes in posture and position of the novel modules not installed conventionally within the building under construction. Therefore, orientational terms must be understood to provide semantic basis for purposes of description, and do not limit the invention or its component parts in any particular way.

The present invention is susceptible to modifications and variations which may be introduced thereto without departing from the inventive concepts. For example,

While the present invention has been described in connection with what is considered the most practical and preferred embodiment, it is to be understood that the present invention is not to be limited to the disclosed arrangements, but is intended to cover various arrangements which are included within the spirit and scope of the broadest possible interpretation of the appended claims so as to encompass all modifications and equivalent arrangements which are possible.

I claim:

1. A method of constructing a frame of a wood frame building structure having a building length axis, the method enabling use of expeditiously fabricated modules, the method comprising:

fabricating wooden frame modules off-site from a construction site including floor modules, ceiling modules, and roof modules, wherein each said module comprises a module length axis, a module width axis, a bounded wood perimeter, and a plurality of ceiling joists, floor joists, or roof rafters oriented in parallel to each other and perpendicularly to the module length axis and coupled to and surrounded by the bounded wood perimeter of each of said floor module, ceiling module, and roof module;

building wall framing, wherein the step of building wall framing comprises building at least a portion of the building wall framing from rectangular wooden frame wall modules each including a bounded wood perimeter and having a longitudinal wall module axis and structural timbers arranged in parallel to each other and perpendicularly to the longitudinal wall module axis and coupled to and surrounded by the bounded wood perimeter;

assembling the frame of the wood frame building structure at the construction site by moving the wooden frame modules into place such that they abut one another, wherein the floor modules, the ceiling modules, and the roof modules are coupled to the building wall framing with respective said module length axes perpendicular to the building length axis, and with said ceiling joists, floor joists, and roof rafters parallel to the building length axis; and

wherein the floor joists of each of said floor modules and the roof rafters of each of said roof modules comprise upper joist boards and lower joist boards parallel to upper joist boards and lower joist boards of every other respective said floor module or roof module, wherein for each of said floor modules and each of said roof modules, a top edge of the upper joist boards is not coplanar to a top edge of the lower joist boards.

2. The method of claim **1**, wherein at least one said wooden frame module includes at least one ledger board.

3. The method of claim **1**, wherein the step of fabricating the wooden frame modules further comprises fabricating the floor modules, the roof modules, the ceiling modules, and the wall modules to be identical in length and width.

4. The method of claim **1**, wherein the step of fabricating the wooden frame modules further comprises fabricating the floor modules, the roof modules, and the ceiling modules to be identical in size and in spacing of floor joists, roof rafters, and ceiling joists.

* * * * *